104A: Midterm 1 Review

Atoms, Orbitals & Periodic Trends

$$E = h\nu = \frac{hc}{\lambda}$$

Bohr [used classical mechanics]

$$E = KE + PE$$

$$= \frac{1}{2} m_e v^2 - k_c \frac{e^2}{r}$$

$$= \frac{Z k_c e^2}{2r} - \frac{k_c e^2}{r} \quad (Z = 1)$$

$$= -\frac{k_c e^2}{2r}$$

Centripetal = Coulombic force

$$\frac{mv^2}{r} = Z k_c \frac{e^2}{r^2}$$

(k_c: Coulomb's constant)

As r decreases, E grows and therefore e^- should just spiral into the nucleus

↳ Bohr proposed quantized angular momentum values:

$$m_e v r = \frac{nh}{2\pi} \quad n = \{1, 2, 3 \dots$$

$$v = n\left(\frac{h}{2\pi}\right)\left(\frac{1}{m_e r}\right)$$

Plugging in

$$\frac{mv^2}{r} = \frac{e^2}{r}$$

to give ⇒

$$r = \frac{n^2 h^2}{4\pi^2 m_e e^2} \Rightarrow E = -\frac{2\pi^2 m_e e^4}{n^2 h^2}$$

These are the possible radi that the e^- can have

* Bohr model of H atom:

$$E = R\left(\frac{1}{n_l^2} - \frac{1}{n_h^2}\right)$$

$$R = \frac{2\pi^2 \mu Z^2 e^4}{(4\pi\varepsilon_0)^2 h^2}$$

where, μ = reduced mass

$$\frac{1}{\mu} = \frac{1}{m_e} + \frac{1}{m_{nucleus}}$$

+ There are more emission lines than absorption

+ $R_H = 2.179 \times 10^{-18} J = 13.6 eV$!

+ $h = 6.3 \times 10^{-34} J_s$

$c = 3 \times 10^8 m/s$

* De Broglie

$$\lambda = \frac{h}{mv}$$

Because e^- are waves, → uncertainties

$$\Delta x \, \Delta p_x \geq \frac{h}{4\pi}$$

→ Heisenberg's uncertainty.

Lymann ($n \to n=1$)

Balmer ($n \to n=2$)

Paschen ($n \to n=3$)

Zeeman effect:

Effect of splitting of a spectral line into several components in the presence of a static magnetic field.

Quantum No's: → These arise from restrictions imposed by Schrod. solutions

All combinations for $n=4$

$n=4$ [Shell]

	$l=0$	$l=1$	$l=2$	$l=3$ [Subshells]
				↓
Total Combinations $= \boxed{2n^2}$ $= 32$	$m_l = 0$	$m_l = 1, 0, -1$	$m_l = 2, 1, 0, -1, -2$	$m_l = 3, 2, 1, 0, -1, -2, -3$
	$m_s = +\frac{1}{2}, -\frac{1}{2}$	$m_s = +\frac{1}{2}, -\frac{1}{2}$	$m_s = +\frac{1}{2}, -\frac{1}{2}$	$m_s = +\frac{1}{2}, -\frac{1}{2}$

ψ → describes motion of e^- in orbital

ψ^2 → probability density of the e^- at any point in space

Pauli Exclusion

+ No two e^- in an atom can have the same set of 4 quantum numbers

Hund's Rule

+ Degenerate orbitals → Max no. of parallel spins results in the lowest $e^- - e^-$ repulsion.

Q] Why does 2s have less energy than 2p?

+ 2s & 2p are both shielded by the $1s^2$ e^-
+ However, 2s has a larger probability density very close to the nucleus.
  2s "penetrates" better than 2p orbital
  ∴ is more stabilized.

Q] From experiments, IE($4s^1$) = 6.62 eV & IE($3d^1$) = 7.98 eV for Sc atom which means 3d is more stable than 4s.

Then why is 4s orbital occupied in ground state?

↓

The Z_{eff} for $4s e^-$ is lower than Z_{eff} for $3d e^-$ (therefore less energy required to remove the 4s e^- than that of 3d)

4s penetrates more
↳ is more stable
∆ fills first
↓
4s extends out more
∆ is shielded by d-orbitals
∆ is therefore ionized first.

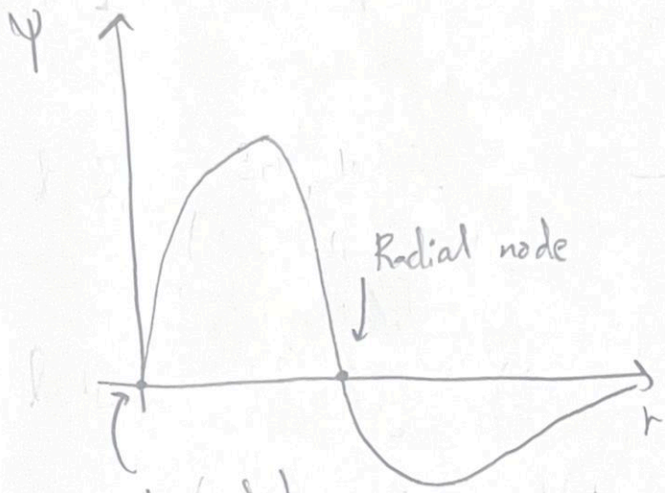
Contour plots → Slice in the given plane

$N = n - 1$ (Total no. of Nodes)

$l = \{0, 1, 2, 3, 4 \dots\}$

s p d f g

l gives the total no. of angular nodes (planar)



→ This is wave function distribution that shows the distribution of e^- radially

↓

Only care about radial nodes.

Not a node! (No sign change)

A node is defined as $\Psi = 0$ and there is a change in sign.

$$\frac{n!}{e!\,(n-e)!}$$

$l = 2$

$3d_{x^2-y^2}$

$N = 2$ → 0 radial, → 2 angular

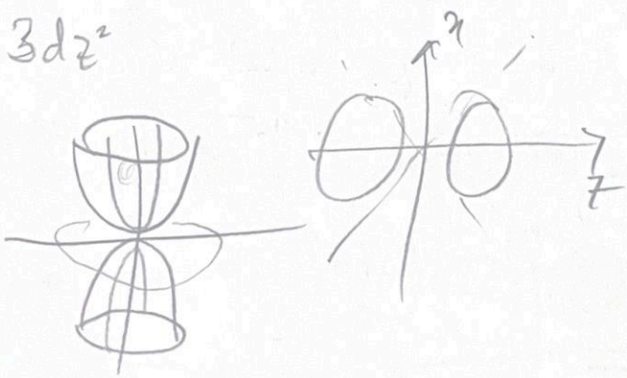
$l = 1$ → Ang

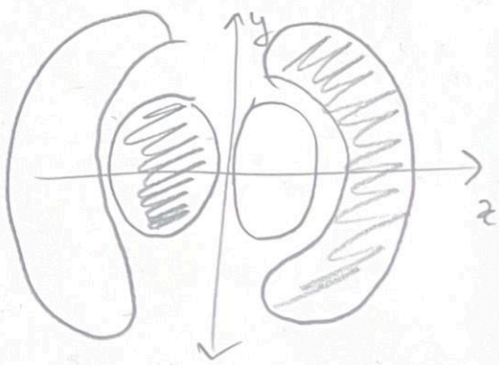
$3p$ → $n = 3$

$N = 2$

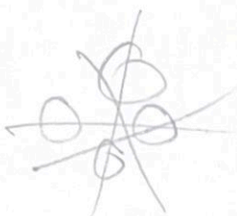
$r = 1$

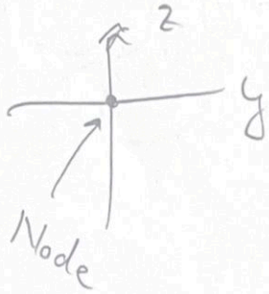
$3d_{z^2}$

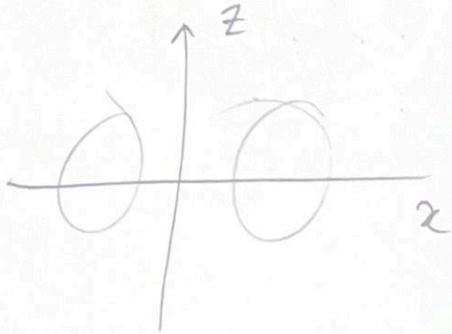


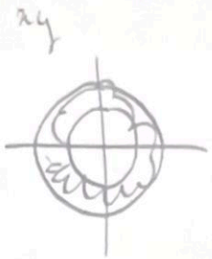


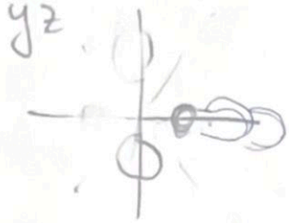


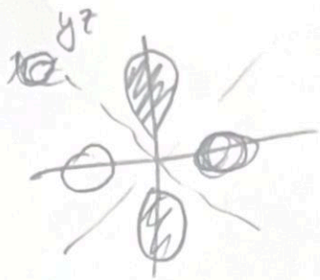


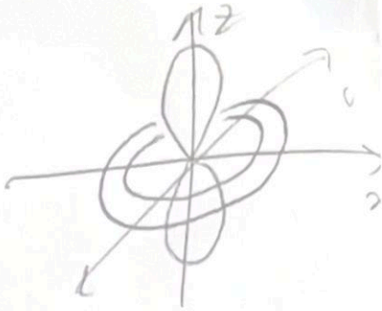




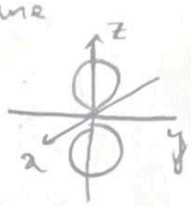


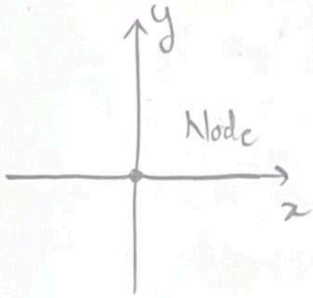




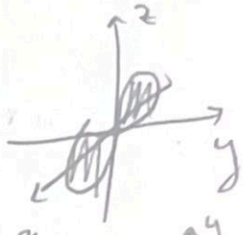


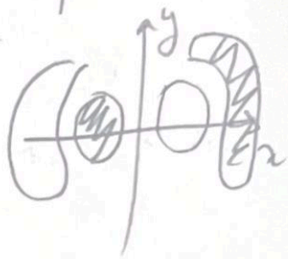
$5p_z \rightarrow$ xy plane





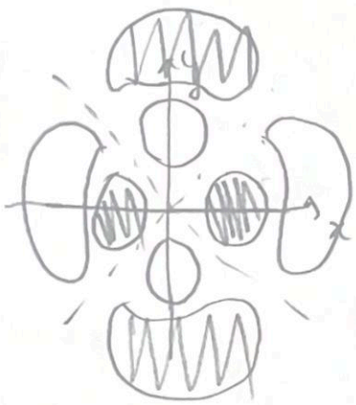
$3p_x$



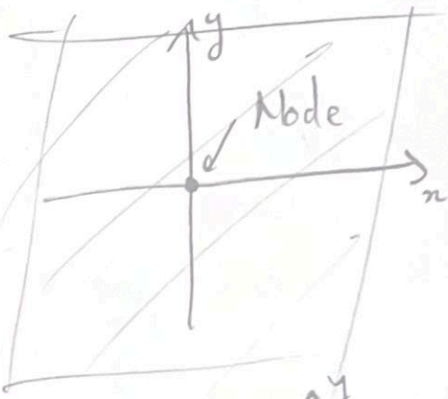


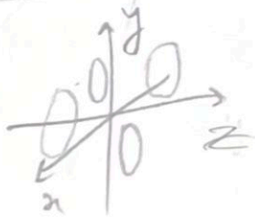
$4d_{x^2-y^2}$

3 nodes
↙ ↘ 1 radial
2 angular

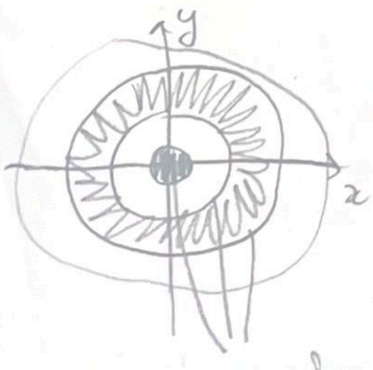


$4d_{xz}$



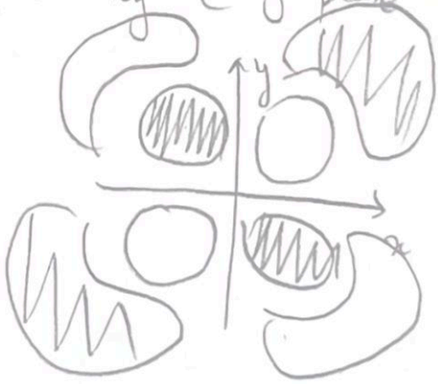


$4s$ (x-y) → 3 nodes

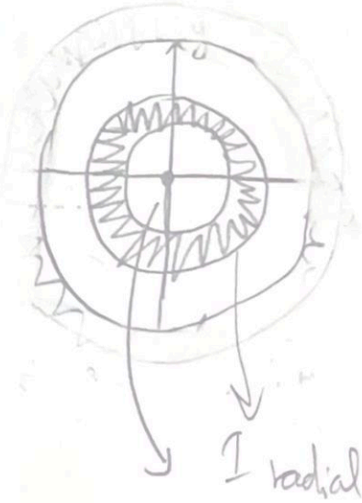


3 nodes

$4d_{xy}$ (x-y) plane



$4d_{z^2}$ → xy plane



1 radial.
ang

er's rule

+ Starting from 1s → clump s and p together, leaving out d and f.

① (1s) (2s, 2p) (3s, 3p) (3d) (4s, 4p) (4d) (4f) (5s, 5p) .....

② Ignore higher groups

(Use total no. of e⁻)

③ (i) Shielding by s & p & (d & f)

- Same bracket → 0.35 (1s → 0.3)
- Net group (to left) → 0.85
- All the others → 1.00

} for d is just 1 for the rest. (previous group is not considered)

eg:

① O: $(1s^2)(2s^2\ 2p^4)$ 2p e⁻ → Z_{eff} = ?

$\sigma = (5(0.35) + 2(0.3))$

$= 1.75 + 0.6 = \boxed{2.35}$ → only when considering the same group.

35 × 5 = 175

$Z_{eff} = 8$

② Fe: $(1s^2)(2s^2\ 2p^6)(3s^2\ 3p^6)(3d^6)(4s^2)$ Z_{eff} (4s e⁻) = ?

$\sigma = (0.35) + (0.85(6) + 18(1))$ → You have to do (n-1)
↳ using all the e⁻ in n=3 (which is 14)

③ Li: $1s^2\ 2s^1$

2s¹: $Z_{eff} = 3 - [2(0.85)]$

1s¹: $Z_{eff} = 3 - [0.3]$

74 - 6 = 68 $5s^2\ 5p^6\ 5d^4$

④ W: [Xe] $(5d^4)(6s^2)$

6s¹: $Z_{eff} = 74 - [(0.35) + 12(0.85) + 60(1)]$

5d¹: $Z_{eff} = 74 - [3(0.35) + 68(1)]$

(4) Mo : $[Kr](4d^4 5s^2)$ $4s^2 4p^6 4d^4$ $42 - 14 = 28$

5s' : $42 - [0.35 + 12(0.85) + 28]$

4d' : $42 - [3(0.35) + 36(1)]$

Lewis + VSEPR + Symmetry } Don't look at double bond or single bonds
↳ just the no. of lone pairs & how many atoms its bonded too

C → rotation
σ → mirror plane
i → inversion
S → rotation + mirror (⊥)

E⁻ domains	0 lone pair	1 lone pair	2	3	4
2	Linear (180) :Ö=C=Ö:				
3	Trigonal planar	Bent <120°			
4	Tetrahedral	Trigonal pyramid	Bent		
5	Trigonal Bipyramidal	Seesaw	T-shape	Linear	
6	Octahedral	Square Pyramid	Square planar	T-shape	Linear

Bent's Rule:

More electronegative substituents prefer site which has larger p character (or less character)

↳ There is more p character in σ bond leading to better overlap & lower energy (greater stability)

+ Higher Z_{eff} → more stable

+ Electronegativities:

(1) Pauling → bond strengths (experimental)

(2) Allred-Rochow

$$\frac{Z_{eff}}{r^2}$$

(3) Mulliken: $X_n \propto \frac{IE + EA}{2}$

+ Fraction of ionic character

$$1 - e^{-1/[4(X_A - X_B)]}$$

Point group table

→ Low order symmetry:

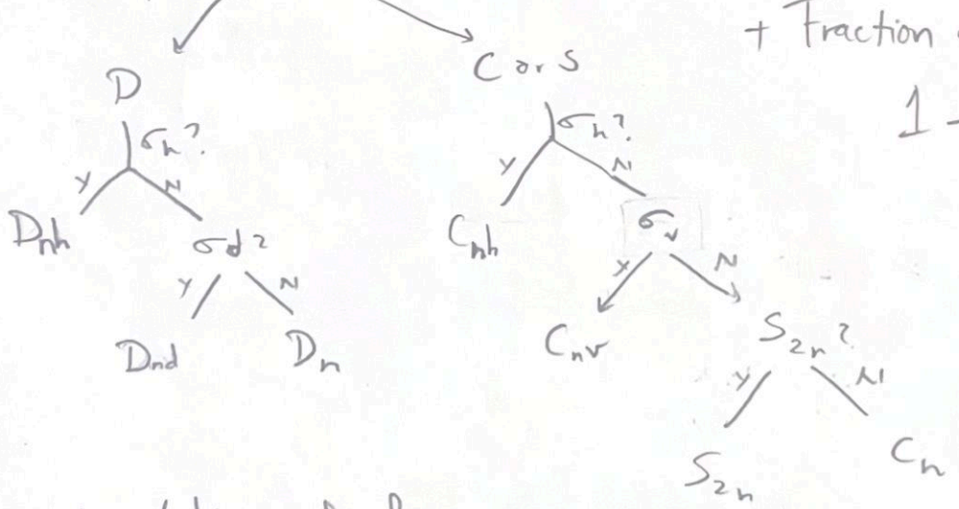
C_1, C_s, C_i

↓

→ High order symmetry

T_d, O_h, ~~$C_{\infty h}$~~, $C_{\infty v}$, $D_{\infty h}$, I_h

→ C_2 perpendicular to principle axis?



dihedral / diagonal plane

↳ contains a principal C_n axis and bisects the angle between two horizontal C_2's (perp. C_2 axis)

If n is odd.

D_{nd} → { E, C_n, nC_2, S_{2n}, i, nσ_d }

D_{nh} → { E, C_n, nC_2, σ_h, i, nσ_v }

(If n is even)

Chem 104A

Normal Quantum no.: n, l, m_l, m_s

- n: principal quantum no. $[1, \infty]$
- l: Orbital angular momentum $[0, \cdots (n-1)]$
- m_l: magnetic quantum no. $[l \cdots -l]$
- m_s: magnetic spin quantum no. $(+1/2, -1/2)$

m_l: +1 0 −1 (1 1 —)

Maximum m_l values when the e^- have been put in subshells → across all the orbitals.

There are existence of microstates due to the interactions between e^- → Not all states are degenerate

Atomic Term Symbols: L, M_L, M_S, S

- L: Maximum M_L value (at the given arrangement of orbitals)
- M_L: M_L of ground state differs from L
- M_S: Max. total spin possible across all the orbitals
- S: Maximum M_S value

$(l=2)$ d 1 1 1 — —

m_l: 2 1 0 −1 2

$M_S = \{3/2, 1/2, -1/2, -3/2\}$

$M_L = \{5, 4, 3, 2, 1, 0, -1, -2, -3, -4, -5\}$

$\boxed{M} = 2S+1$ ← Multiplicity

→ $J = [L+S, |L-S|]$ ← total angular momentum (spin-orbit coupling)

If $|L+S| = |L-S|$ only one J value exists

If not then increments of 1

M_L, M_S → describe collections of microstates
S, L → describe the microstates themselves

} L & S are the lowest possible values of M_L & M_S

L = 0 → S
L = 1 → P
L = 2 → D
L = 3 → F
L = 4 → G
L = 5 → H

(Similar to s, p, d, f but not the same)

(Note: S (L=0) and S (atomic spin quantum no. are the same symbol!)

Now a term symbol is: + a value of L (S, P, D, F)
+ left superscript: M

M = 3

$^3D_2$ → (L=2)

→ J (max J for more than half filled & min J for less than half filled)

Ground Term:

1. Has highest spin multiplicity
2. If 1 is same, then highest L value

S, P, D, F, G, H, I

Solving a problem:

$\frac{1}{+1}\ \frac{1}{0}\ \frac{\ }{-1}$

+ If asking for ground state then have to fill in such that m_s is maximised
+ If not then we can flip and change the e^- to maximise certain configurations to get all possible M_S & M_L values.

+ Then you can fill in the microstate tables by playing w/ electron combinations until they add up to the required no. AUFBAU PRINCIPLE

(MAKE SURE to not break

Bohr | $E = h\nu = hc$. $E = KE + \Phi$. [illegible]

Final Review: Chem 104A

Ashutosh B.

Acids & Bases:

+ **Brønsted-Lowry:** Acid: species w/ a tendency to lose a hydrogen ion
  Base: species w/ a tendency " gain a hydrogen ion

As the strength depends on the solution, BL strategy is to compare conjugate acids and bases.

→ **Superacids:** Acids that are more acidic than sulphuric acid. (exceedingly weak conjugate bases have)

↳ Superelectrophilic activation [addition of large amount of positive charge]

Hammett acidity: $H_0 = pK_{BH^+} - \log \frac{[BH^+]}{[B]}$ B → nitroaniline indicator; BH^+ → conjugate base

+ **Hess's Law:** Instead of measuring the enthalpy change of just deionize completely → we divide the reaction into steps that go to completion

eg: HA is a weak acid (to find its enthalpy of ionization, we will follow these steps)

(Reaction w/ strong base) $HA(aq) + OH^-(aq) \longrightarrow A^-(aq) + H_2O(l)$ ΔH_1

(-1) $H_3O^+(aq) + OH^-(aq) \longrightarrow 2H_2O(l)$ ΔH_2

$HA + H_2O \longrightarrow H_3O^+ + A^-$ $\Delta H_3 = \Delta H_1 - \Delta H_2$

+ **Van't Hoff:** Another method is to use K_a vs. T to construct:

$$\ln K_a = \frac{-\Delta H_3}{RT} + \frac{\Delta S_2}{R}$$

Gas-phase vs. Solution phase:
↳ easier to measure as solvent effects are not applicable [Gas-phase Acidity, Gas-phase Basicity]

Superbases: Specis that have gas-phase proton affinities (PA) > 1000 kJ/mol

Trends: Its imp to know how to rank the specis based on higher & lower strengths

[Note: there is a somewhat correlation between gas-phase and aqueous but w/ exceptions]

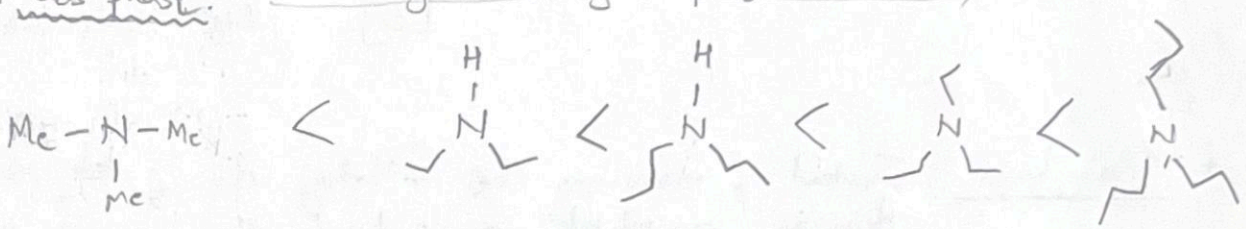
① **Inductive effects** → Alkyl Groups are e^- donative ⇔ create a more e^- rich basic atom centers increasing BL base strength

→ highly electronegative atoms/gases (Br, Cl, F, CF_3, CF_3SO_2) draw e^- density away from Brønsted basic atom

→ a larger alkyl chain further enhances the effect

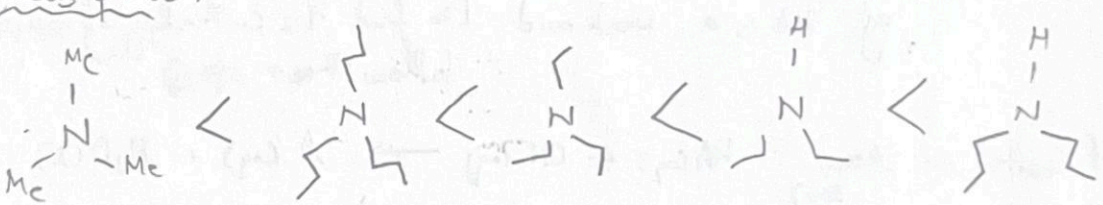
eg: Gas phase:

greater strength base →



Exception! "Aqueous phase":

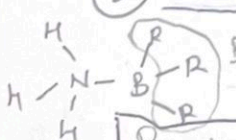
better base →



These are diff from gas phase because tertiary amines have lower solvation of protonated cations. This is because, H atoms are needed for hydrogen bonding (highly substituted are less basic) ★

Therefore, NH_3 is a very strong aqueous base but a very weak gas phase base!

② **Steric effects**: Steric effects make the basic atom less accessible and more difficult to solvate upon protonation



B-strain vs. F-strain (large groups on other species)

B-strain: F-strain but due to large groups on one species

F-strain: non-bonded repulsion of bulky groups → Make it difficult for Lewis acid & Lewis base centers to approach & interact.

③ Conjugate base. Many acid strengths can be predicted by analyzing the stability stability of conjugate base.

Going down group increases acidity. { Acidity $\propto$ No. of e^- in central atom ; Acidity $\propto \downarrow$ electronegativity

$H_2Se > H_2S > H_2O$

$SeH^- \quad SH^- \quad OH^-$

} Large, heavy species can better stabilize the negative charge and have smaller attraction of H

Across a period: Acidity $\propto \uparrow$ electronegativity

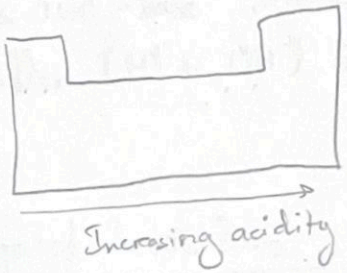
$NH_3 < H_2O < HF$

$NH_2^- \quad OH^- \quad F^-$

} strength decreases towards left in the periodic table.

↓

But overall it is still not very acidic due to the stability of conjugate base



∴ HI → very strong acid

④ Resonance: The stability of conjugate base has also got to do of the resonance stabilization of the conjugase base

↓

The more the negative charge can delocalize, the weaker the conjugate base and therefore the stronger the acid. (less more stable)

eg: Oxygacids

$H-O-Cl < H-\overset{\overset{O}{|}}{O}-Cl < H-O-\overset{\overset{O}{|}}{Cl}-O < H-O-\overset{\overset{O}{|}}{Cl}-O$

electrostatic parameter

⑦ Metal cations: Metal cations are positive and therefore draw the bonding electron density towards them - weakening the OH bond water that is bound to them → resulting in a bound OH while releasing H^+ in the solution

$\xi = \frac{z^2}{d}$ (z^2 → charge; d → M^{n+} radius + H_2O diameter)

Smaller radii and large positive charged ions are stronger acids!

Lewis!

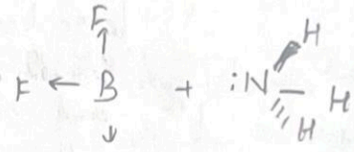
(e^- pair acceptor) (e^- pair donor)

Lewis Acid + Lewis base = Adduct [Coordination compound]

(This special name because the bonding is actually coordinate covalent bonding!)

eg: $BF_3 + NH_3$

↓ electron deficient due to highly e^- F

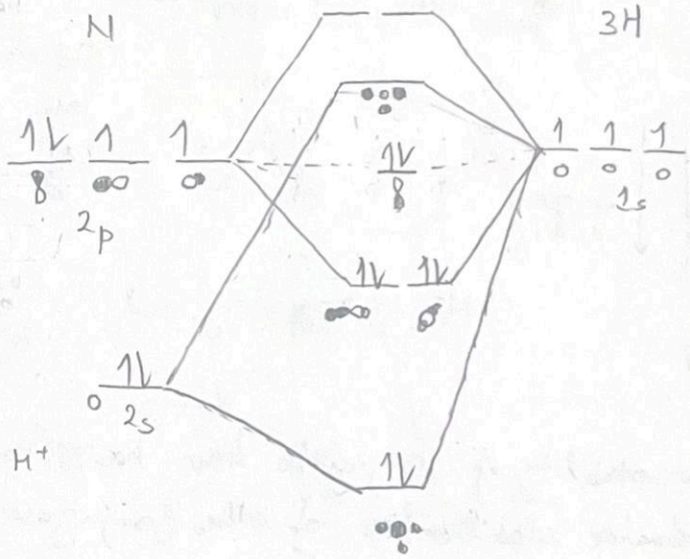


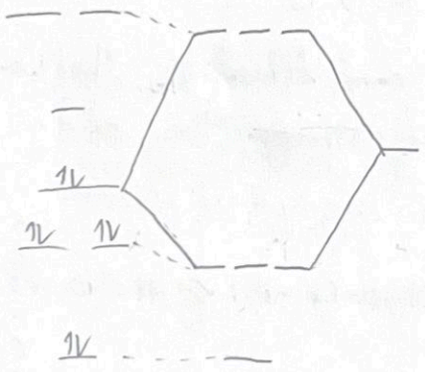
LUMO

HOMO (Because e^- come from base, they have to be in an occupied orbital)

These are the 'frontier' orbitals that take part in forming new HOMO & LUMO orbitals of the product.

eg: $NH_3 + H^+ \longrightarrow NH_4^+$





-rough we can use organic chemistry arguments to find acid, base strengths.
There are a few exceptions;

Boron halide Lewis acid strength:

$BI_3 > BBr_3 > BCl_3 > BF_3$!!

Although F is more electronegative and manages to pull e^- towards itself, the B-X bond is an adduct! (coordination bond)

:X: donate a lone pair to B filling its p orbitals.

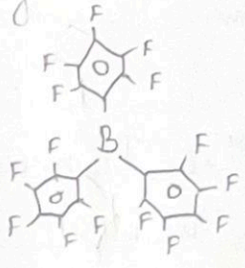
More bigger the halide, the worse the overlap and therefore the more e^- deffiaent the B atom (since e^- can't be shared properly by the halides)

The trends discussed earlier are Lewis acid-base interactions w/ smaller (H^+ or OH^-) ions

This order significantly changes when the interacting species is changed to BF_3 or BMe_3 as steric effects (F strain & B-strain) are taken into account.

Frustrated Lewis Pairs

Combination of Lewis acids and bases that can have strong interaction but "frustrated" by strong steric hindrance. → These then possess unusual reactivity.



tris(pentafluorophenyl)borane



What happens?
- Nucleophilic attack
- Forming a zwitterion

This P is made "frustrated" as a small activation energy is built up

Hard-Soft Acids & Bases [HSAB]

★ Polarizability: Acids & bases that are polarizable (i.e. the electron density is such that e^- in polarizable molecules can be attracted or repelled by charges on other molecules.)

⇓

These are termed SOFT

Hardness or softness has no correlation w/ strength!

Hard likes hard; Soft likes soft

(ionic interaction)

→ Cations w/ +3
→ Alkaline earth

Exceptions: Cr^{3+}, Mn^{2+}, Fe^{3+}, Co^{2+}

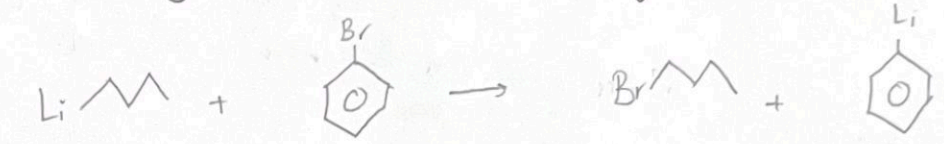
(covalent interaction)

+ d e- are readily available for π bonding
+ large, and heavy atoms

[More favorable reactions that form more stable products]

Hardness/Softness and strength are defined independtly and used together to describe

eg:



hard-soft

hard-soft

?

Products are favored because

Br is more basic than

Quantitative measures:

$$\eta = \frac{IE - EA}{2}$$; $$\chi = \frac{IE + EA}{2}$$

absolute hardness

absolute electronegativity

These assumes that IE measures E_{HOMO} and EA measures the LUMO.

∴ A higher η means harder

χ ↑ means "

$$\sigma = \frac{1}{\eta}$$

Softness

[More polarizable will have ↑C_A → wants to form covalent bonds!]

Drago and Wayland

Acids have low C_A & high E_A

$$-\Delta H = E_A E_B + C_A C_B$$

Capacity for electrostatic interactions

Tendency to form covalent bonds

A → Acid
B → Base

Midterm 2 Review

$E: \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

1) Matrix: Satisfy properties of a point group → Represent an operation on the group
(Matrix representation)

2) Characters: Trace of a matrix (i.e. sum of no. on diagonal)

↳ Reducible representation (Γ) } Character table

↳ Irreducible representation (A_1, B_{2u}, A_{2g}) ⟶ Derived from mulliken symbols.

3) Order: Total no. of symmetry operations (including the no. of these that exist more than 1 time)
(h)

4) Classes: No. of columns in character table

No. of irreducible reps. = Columns [Character table is always Square]

Some properties:

(i) $h = \sum [\text{Dimension}(E)]^2$

eg: $h = 1^2 + (-1)^2 + 1^2 = 3$

(ii) $h = \sum \left[\text{Dimension} \cdot \text{No. of operation in class} \right]^2$

(iii) Orthogonality: $h = \sum \left[\text{Dim} \cdot \text{No. of operation in class} \right] = 0$ (Not squared!)

(iv) All groups have a totally symmetric represention that has characters of 1 for all operations.

Point group → C_{2v}	E	C_2	σ_v	Σ = order
Irreducible reps: A_1	1	0		
A_2	-1	1		
A_3	1	1		

Operation; Classes

(X) Dimensions / characters (from matrix representation)

Using Symmetry

+ Vibrations

	Vibrational Modes	(x, y, z)
Linear:	3N-5	(3 translational + 2 rotation)
Non linear:	3N-6	(3 trans + 3 rot.)

Post MT2 Shit

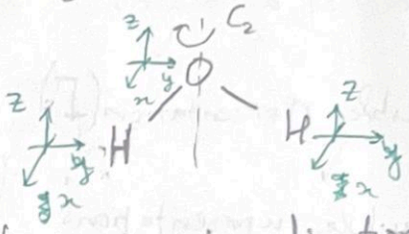
Rules:

1) If atom changes position → 0
2) If " remains but one vector is reversed → -1
3) If " remains and no direction change → $+1$

Use this on each atom to find the reducible representation (Γ)

eg:

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v'(yz)$	
Γ	3	-1	-1	-3	α



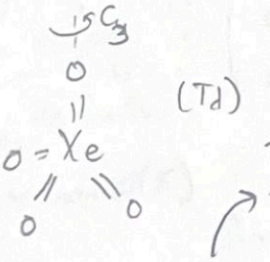
You have to do for every vector direction i.e. if two vector directions change then it should be -2.

	E	C_2	$\sigma_v(xz)$	$\sigma_v'(yz)$	
$\therefore \Gamma_{\text{stretch}} =$	9	-2	-1	-3	??? (Is this correct?)

However,

We can also specifically focus on certain bonds to figure out stretching modes. In this case we have only 1 axis (the bond axis) and the rules become simpler: $+1$: If bond vector is unchanged

0 : If bond vector moves.

eg: 

	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	4	1	0	0	2

This is specific

Γ_{stretch} for Xe–O bond.

MO's, Hybridisation & VB Theory

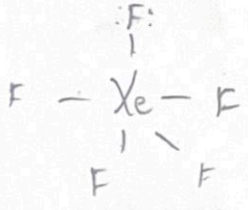
1) Hybridisation : s, p, d can all hybridise to give equal & unequal MO's.

↳ You just count the no. of bonds being formed, add up orbitals to reach the bond size. 4 bonds → sp^3 (Tetrahedral) – equal orbitals

5 bonds → sp^3d (trigonal bipyramida) → unequal orbita

VSEPR : Repulsion order : LP-LP > LP-BP > BP-BP

eg: XeF_5^- [e^- : 8+35+1 = 44e^-]

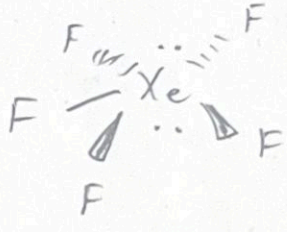


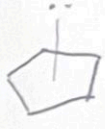
3(5) = 30 e^-

5 bonds = 10 e^-

= 40 e

2 lone pairs





pentagonal bipyramidal

hybridisation : sp^3d^3

4 3 → 7 bonds (including lone pairs)

2) Valence Bond Theory

+ Overlap of half-filled valence AO's makes a covalent bond
+ σ & π bonds
+ Hybridisation occurs to form bonds.

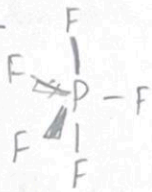
Chem 104A: Review (CCC Tutoring)

↑S-p mixing

σ_g (from 2s) and σ_g from 2p are close (B_2, C_2 & N_2) in the MO for these 3 molecules and that is why there is s-p mixing.

MO diagrams

(1) PF_5

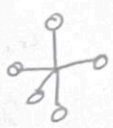


D_{3h}

35 valence e^- + 5 ⇒ 40 e^-

1) Draw Lewis Structure
2) Point group
3) Construct SALC's (from symmetry table)

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ_{5F}	5	2	1	3	0	3

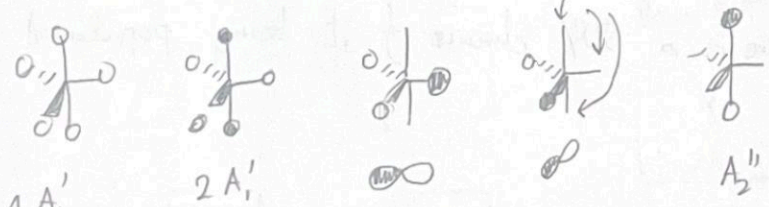


This is the base case we have

(irreducible) $\Gamma_{5F} = 2A_1' + E' + A_2''$ } These should add up to reducible

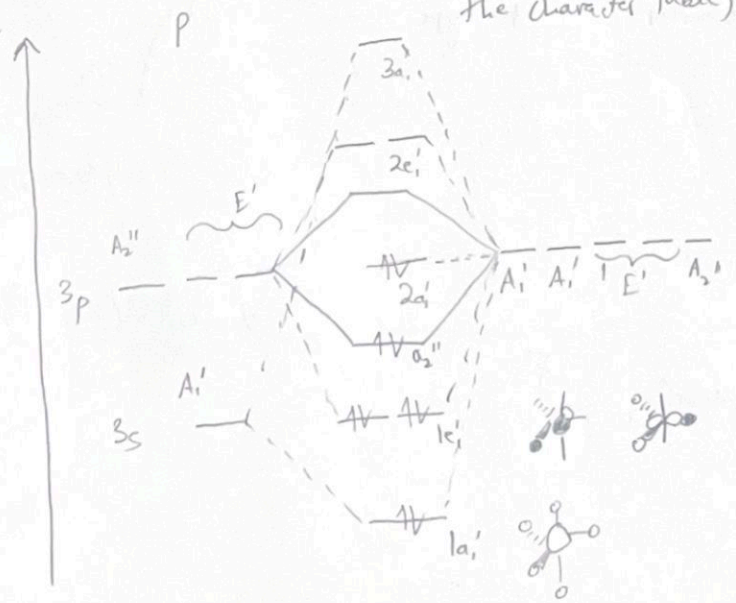
↓

Based on these → you use Mulliken symbols to design the shape of the SALC



E' (we know (x,y) from the character table) (because of symmetry to $p_{x,y}$)

Now P: 3s ↓ A_1' ; 3p ↘ E' & A_2''





Bond order:

$\frac{8e^-}{2}$ (nb does not count)!

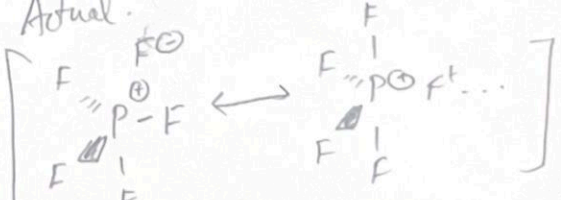
= 4 bonds

↓

spread across 5 bonds

0.8 bond order

Actual:



Drago-Wayland: $-\Delta H = E_A E_B + C_A C_B$

1-3
$-\Delta H = E_1 E_3 + C_1 C_3$
$= 16.3 + 15.9$
$= 32.7$

1-4
$-\Delta H = E_1 E_4 + C_1 C_4$
$= 23 + 4.95$
$= 27.95$

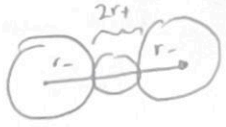
2-3
$-\Delta H = 13.2 + 9.77$
$= 22.97$

2-4
$-\Delta H = 18.1 + 3.05$
$= 21.15$

1-2
$-\Delta H = 225 +$

Because it is 1 equivalent. The larger sum is favored.

BCC O_h sites



$2r_+ + 2r_- = a$

a for BCC: a, $a\sqrt{2}$, $a^2 + 2a^2 = a\sqrt{3}$

$4R = a\sqrt{3}$

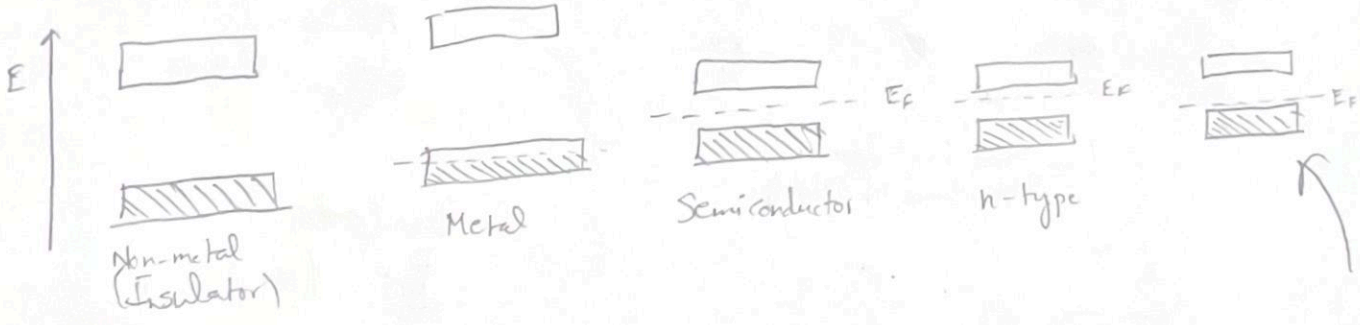
$R = \frac{a\sqrt{3}}{4}$

$2r_+ = \frac{4R}{\sqrt{3}} - 2R$

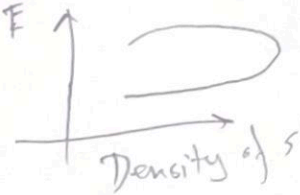
$r_+ = \frac{4R - 2\sqrt{3}R}{2\sqrt{3}}$

$= \frac{2R - \sqrt{3}R}{\sqrt{3}} \approx \boxed{\frac{2R}{\sqrt{3}} - R}$

Fermi-level: energy level @ which there is a 50% chance of it being populated



Draw w/ Parabolas!



Vincent (CL7)

Vibration Γ_{xyz} (from character table), $\Gamma_{unmoved}$, Γ

Post MT2 Shit

Peric 1A Recap:

Hard vs. soft

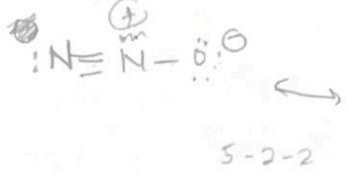
↓

Usually first row

↓

smaller radii elements are less polarizable so hard

Identify lewis acid base from lewis structure



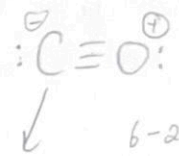
10+6 = 16

5-2-2

Hard acid + Hard base → More stability (electrostatic effect → more coulombic interaction } leaning towards ionic

Hard soft (More covalent)

soft soft



6+4

6-2-3

More negative so a larger e^- cloud which makes it ~~more~~ easily polarizable

↓

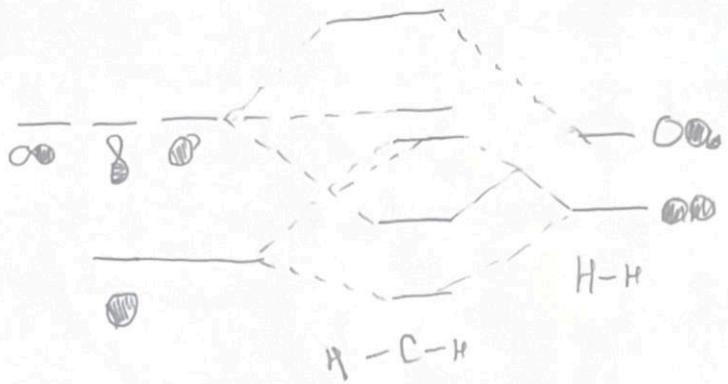
soft

So generally, more electronegative atoms want to hold e^- close to themselves and are less polarizable

↓

Harder

H–C–H



Book:

Albright, Burdett + Whangbo (Chapter 7)

You have to check the orthogonality of each orbital interaction:



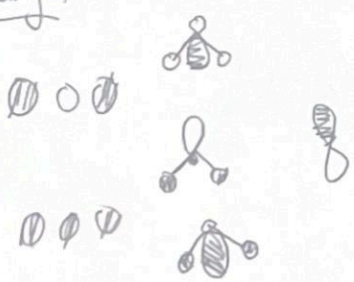
$\begin{pmatrix} C \\ 1s \end{pmatrix}$ + antibonding H are orthogonal as any overlap is cancelled by forming a bond and then an antibond.

Similarly



} No interaction and therefore will not interact in the MO diagram.

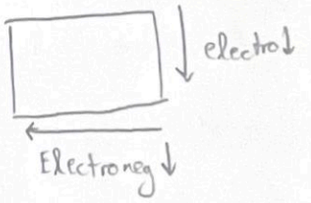
S-p mixing:



Post MT2 Shit

Periodic Table Stuff

Group I → Alkali metals, II → Alkali earth, III → Triel elements, IV → Tetrel elements
V → pnicogens; VI → Chalcogens; VII → Halogens; VIII → Noble gases.



IA follow a similar trend w/ a few exceptions'
(Boron, Oxygen)
They become completely/half filled after losing one e⁻ → Therefore want to readily achieve that achieve greatest exchange energy.

↳ Hydrogen ($1s^1$)
+ Most abundant element in the universe
+ Protium, deuterium, tritium
+ Fuel potential but hard to produce & store
+ H^- → has noble gas configuration termed as a hydride
↳ Powerful reducing agents (BH_4^-, AlH_4^-)

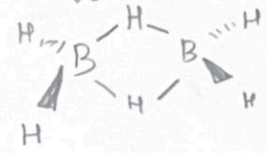
↳ Cracking hydrocarbons:
$C_2H_6 \rightarrow C_2H_4 + H_2$
↳ Steam re-forming
$CH_4 + H_2O \rightarrow CO + 3H_2$

↳ Boron ($1s^2 2s^2 2p^1$)
+ Although it behaves as a non-metal, its chemistry is very different from its group. 13 (Similar to that of C and Si)
+ Forms hydrides (boranes) and oxygen-containing minerals (borates)
+ Many allotropes (based on icosahedral B_{12} unit) → I_h symmetry

↳ Boranes: Three-center, two-electron bonding (involves 3 atoms and two bonding e⁻ per bridge)

Diborane

D_{2h} Symm.



We can use MO theory to help build and predict the bonding in B_2H_6



The p orbitals are involved w/ bonding w/ bridge H'



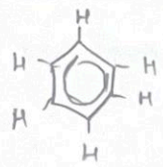
These are calculated using the rules:
+1 for no change
-1 for flip

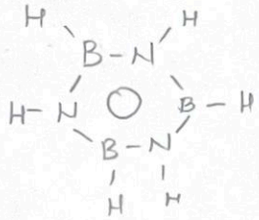
	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$
$\Gamma(p_z)$	2	2	0	0	0	0	2	2
$\Gamma(p_x)$	2	-2	0	2	0	0	2	-2
$A_g + B_{3u}$ Γ_{1s}	2	0	0	2	0	2	2	0

Carboranes: Contain both boron and carbon atoms arranged in clusters. [ortho, meta and para carboranes]

BX_3 → are Lewis acids (accept an e^- pair from a halide to form BX_4^-)

Benzene vs. **Borazine** ($B_3N_3H_6$)





Similar: Structure/Shape, density, surface tension

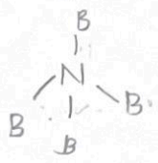
Borazine is readily formed by reaction of B_2H_6 and NH_3:

$$\tfrac{3}{2} B_2H_6 + 3NH_3 \longrightarrow B_3N_3H_6$$

Different: melting, boiling point

B–N bonds: BN exists in Zinc Blende Structure

- **diamond-like form** → N Tetrahedrally to 4 B (and vice-versa)



- **graphite-like form**
  - + BN occurs in extended fused-ring systems
  - + Much less delocalization of pi e- (making it a poor conductor)
  - + white, insulator and can be used as lubricant.

Shapes of Borohydrides



Closo $(B_6H_6)^{2-}$ → nido (B_5H_9) → arachno (B_4H_{10})

Usually the symmetry elements range from $C_s \rightarrow C_{4v}$